Suyoung Lee

Curriculum Vitae

Contact Information

Affiliation:	Department of ECE, ASRI, Seoul National University, Seoul, Korea
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Research Interests

My research focuses on deep learning for computer vision and graphics, with a particular emphasis on 3D scene understanding and generation. I am currently working on accurate and high-quality 3D reconstruction and large-scale scene generation, with a growing interest in developing scalable, efficient 3D scene generation frameworks. In particular, I focus on holistic scene reconstruction from omnidirectional images, a capability that directly translates to practical applications such as digital twin construction of manufacturing facilities and immersive XR experiences, where rendering pipelines must run reliably under real system constraints.

My earlier work in low-level vision—including video restoration and frame interpolation—continues to inform my approach to producing efficient, high-fidelity visual outputs and has been extended to semiconductor structures analysis through an industry collaboration on FIB-SEM tomography interpolation. Overall, I aim to develop 3D reconstruction and generation techniques that bridge benchmark-level performance and real-world industrial deployment, spanning scales from nanometer-level structural inspection to facility-level spatial modeling.

Education

Seoul National University (SNU), Seoul, Korea Mar. 2019 -
Integrated M.Sc./Ph.D. course in Electric and Computer Engineering
GPA: 4.27 / 4.30 (60 credits in Ph.D. Course of Electrical and Computer Engineering)
Advisor: Prof. Kyoung Mu Lee

Seoul National University (SNU), Seoul, Korea Mar. 2013 - Feb. 2019
B.Sc. in Electric and Computer Engineering
Summa cum laude, GPA: 4.18/4.30 (136 credits in Electrical and Computer Engineering)
Leave for military service (Korea Airforce): Mar. 2015 - Mar. 2017 (24 months)

Gyeonggi Science High School, Suwon, Korea Mar. 2010 - Feb. 2013

Publications

- **Suyoung Lee**, Jaeyoung Chung, and Kyoung Mu Lee, "OmniAdapt: Fast Test-Time Adaptation of Feed-Forward 3D Gaussian Splatting from Omnidirectional Images" (Under review).

- Jaeyoung Chung*, **Suyoung Lee***, Jianfeng Xiang, Jialong Yang, and Kyoung Mu Lee, "Map2World: Segment Map Conditioned Text to 3D World Generation," in Proceedings of European Conference on Computer Vision (**ECCV**), 2026.[1]
  works done during internship at Microsoft Research

- Sangmin Hong*, **Suyoung Lee***, and Kyoung Mu Lee, "StarryGazer: Leveraging Monocular Depth Estimation Models for Domain-Agnostic Single Depth Image Completion," to appear in Signal, Image and Video Processing (**SIVP**), Springer, 2026.

- Jaeyoung Chung*, **Suyoung Lee***, Hyeongjin Nam, Jaerin Lee, and Kyoung Mu Lee, "LucidDreamer: Domain-free Generation of 3D Gaussian Splatting Scenes," in IEEE Transactions on Visualization and Computer Graphics (**TVCG**) **Runner-Up of the 2025 Best Paper Award in TVCG Computer Graphics Area**, 2025 (250+ Citations, 1,500+ GitHub Stars).

- **Suyoung Lee***, Jaeyoung Chung*, Kihoon Kim, Jaeyoo Huh, Gunhee Lee, Minsoo Lee, and Kyoung Mu Lee, "OmniSplat: Taming Feed-Forward 3D Gaussian Splatting for Omnidirectional Images with Editable Capabilities," in Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition(**CVPR**) **Highlight**, 2025.

- **Suyoung Lee***, Jaeyoung Chung*, Jaeyoo Huh, and Kyoung Mu Lee, "ODGS: 3D Scene Reconstruction from Omnidirectional Images with 3D Gaussian Splatting," in Proceedings of Neural Information Processing Systems (**NeurIPS**), 2024.

- Myungsub Choi, **Suyoung Lee**, Heewon Kim, and Kyoung Mu Lee, "Motion-aware dynamic architecture for efficient frame interpolation," in Proceedings of the IEEE/CVF International Conference on Computer Vision (**ICCV**), 2021.

- Sanghyun Son, **Suyoung Lee**, Seungjun Nah, Radu Timofte, and Kyoung Mu Lee, "NTIRE 2021 challenge on video super-resolution," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**) Workshop, 2021.

- Seungjun Nah, Sanghyun Son, **Suyoung Lee**, Radu Timofte, and Kyoung Mu Lee, "NTIRE 2021 challenge on image deblurring," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**) Workshop, 2021.

- **Suyoung Lee***, Myungsub Choi*, and Kyoung Mu Lee, "DynaVSR: Dynamic Adaptive Blind Video Super-Resolution," in Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), 2021.

Experience

• **Research Internship @ Microsoft Research Asia (Beijing)** Jul. 2025 - Dec. 2025
Topic: Large-scale 3D World Generation.
Group: Visual Computing Group.
Mentor: Jialong Yang `https://jlyang.org/`.
Role description: Adapted a pretrained transformer-based 3D foundation model (TRELLIS) to scene-level generation through a parameter-efficient adapter, working alongside the model's original authors.

[1] * indicates equal contribution

• **Industry Collaboration Project with LG AI Research** Nov. 2023 - Oct. 2024
Topic: Development of 3D Scene Reconstruction and Generation Technique for Real-world Applications.

• **Industry Collaboration Project with Mechatronics R&D Center, Samsung Electronics (Current Seminconductor R&D Center and AI Center)** Mar. 2020 - Dec. 2020
Topic: Learning-Based Data Interpolation Algorithm Using Multi-Layer Multi-Wavelength Inspection Images for Semiconductor Structure Analysis.

Service

• **Workshop Challenge Co-organizer @ CVPR 2021** Jun. 2021
NTIRE 2021 Challenge on Image Deblurring and Video Super-Resolution.

• **Reviewer**
IEEE TIP (2025-), IEEE TVCG (2026-).
CVPR (2023-), NeurIPS (2025-), ICCV (2023-), ECCV (2024-), WACV (2023).

Scholarships

• **Electrical Engineering and Computer Science Graduate Student Program**, 2019-2023
Korea Foundation for Advanced Studies

• **National Scholarship for Science & Engineering** 2013-2018
Korea Student Aid Foundation

Skills

Programming: PyTorch, Python (High), C++, CUDA, MATLAB (Low)
Language: Korean (Native), English (Fluent)

References

Advisor Kyoung Mu Lee
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`https://cv.snu.ac.kr/index.php/kmlee`